BIOLOGY OBJECTIVE QUESTIONS

170 MCQs • Correct Answers • Short Explanations • Simple & Twisted Questions

Based on the 11 chapters shown in the supplied Table of Contents. The questions are arranged by chapter and include direct recall, conceptual and exam-style twisted questions.

Coverage

• CHAPTER ONE — PROTOZOA

• CHAPTER TWO — PORIFERA, CNIDARIA & CTENOPHORA

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• CHAPTER TEN — REPTILIA & AVES

• CHAPTER ELEVEN — MAMMALIA

• □ HIGH-YIELD MIXED/TWISTED QUESTIONS

CHAPTER ONE — PROTOZOA

1. Protozoans are generally regarded as:

A. Multicellular prokaryotes
B. Unicellular eukaryotes
C. Multicellular eukaryotes
D. Unicellular prokaryotes

✓ **Correct Answer: B. Unicellular eukaryotes**

Explanation: Protozoans are traditionally described as unicellular eukaryotic organisms; each organism is generally a single cell with a nucleus and membrane-bound organelles.

2. Which of the following is NOT a characteristic of protozoans?

A. They are mostly microscopic
B. They are generally unicellular
C. They possess tissues and organs
D. Many are heterotrophic

✓ **Correct Answer: C. They possess tissues and organs**

Explanation: Protozoans do not form true tissues and organs. Tissue-level organization is a feature of multicellular animals.

3. The locomotory organelle of Amoeba is:

A. Cilia
B. Flagellum
C. Pseudopodium
D. Seta

✓ **Correct Answer: C. Pseudopodium**

Explanation: Amoeba moves by extending temporary cytoplasmic projections called pseudopodia, which also help it engulf food.

4. Pseudopodia are primarily used for:

A. Movement and feeding
B. Respiration only
C. Reproduction only
D. Excretion only

✓ **Correct Answer: A. Movement and feeding**

Explanation: Pseudopodia help Amoeba move and surround particles during phagocytosis.

5. Which organism moves mainly by cilia?

A. Amoeba
B. Paramecium
C. Plasmodium
D. Trypanosoma

✓ **Correct Answer: B. Paramecium**

Explanation: Paramecium has numerous cilia that beat in coordinated waves to produce movement and help create feeding currents.

6. The contractile vacuole in freshwater protozoans is mainly involved in:

A. Digestion
B. Reproduction
C. Osmoregulation
D. Photosynthesis

✓ **Correct Answer: C. Osmoregulation**

Explanation: Freshwater is hypotonic to the cell, so water continually enters by osmosis. The contractile vacuole removes excess water and prevents bursting.

7. The macronucleus of Paramecium is mainly concerned with:

A. Sexual reproduction
B. Vegetative activities
C. Locomotion
D. Excretion

✓ **Correct Answer: B. Vegetative activities**

Explanation: The macronucleus controls routine metabolic and vegetative activities such as growth and cell function.

8. The micronucleus of Paramecium is particularly important in:

A. Conjugation
B. Digestion
C. Locomotion
D. Osmoregulation

✓ **Correct Answer: A. Conjugation**

Explanation: The micronucleus participates in the genetic exchange that occurs during conjugation.

9. Which of the following causes malaria?

A. Trypanosoma
B. Entamoeba
C. Plasmodium
D. Giardia

✓ **Correct Answer: C. Plasmodium**

Explanation: Malaria is caused by protozoan parasites of the genus Plasmodium.

10. The vector of human malaria is the:

A. Male Anopheles mosquito
B. Female Anopheles mosquito
C. Female Culex mosquito
D. Male Aedes mosquito

✓ **Correct Answer: B. Female Anopheles mosquito**

Explanation: The female Anopheles mosquito takes blood meals and transmits Plasmodium; males do not bite humans.

11. The infective stage of Plasmodium transmitted to humans by the mosquito is the:

A. Merozoite
B. Gametocyte
C. Sporozoite
D. Zygote

✓ **Correct Answer: C. Sporozoite**

Explanation: Sporozoites are injected into humans from the mosquito's salivary glands during a bite.

12. Amoebic dysentery is caused by:

A. Entamoeba histolytica
B. Entamoeba coli
C. Plasmodium vivax
D. Giardia intestinalis

✓ **Correct Answer: A. Entamoeba histolytica**

Explanation: Entamoeba histolytica is an intestinal protozoan capable of invading the colon and causing amoebic dysentery.

13. Which protozoan is associated with African sleeping sickness?

A. Trypanosoma
B. Leishmania
C. Paramecium
D. Plasmodium

✓ **Correct Answer: A. Trypanosoma**

Explanation: African trypanosomiasis is caused by Trypanosoma species and is transmitted by tsetse flies.

14. African sleeping sickness is transmitted by:

A. Housefly
B. Tsetse fly
C. Sandfly
D. Blackfly

✓ **Correct Answer: B. Tsetse fly**

Explanation: The tsetse fly is the biological vector that transmits Trypanosoma parasites causing African sleeping sickness.

15. Which of the following protozoans possesses a flagellum?

A. Amoeba
B. Paramecium
C. Trypanosoma
D. Plasmodium

✓ **Correct Answer: C. Trypanosoma**

Explanation: Trypanosoma has a flagellum associated with an undulating membrane, helping it move through body fluids.

16. Twisted: An organism uses temporary cytoplasmic projections for both locomotion and engulfing food. It is most likely:

A. Paramecium
B. Amoeba
C. Trypanosoma
D. Plasmodium

✓ **Correct Answer: B. Amoeba**

Explanation: Temporary cytoplasmic projections are pseudopodia, the characteristic locomotory and feeding structures of Amoeba.

17. Binary fission in protozoans is a form of:

A. Sexual reproduction
B. Asexual reproduction
C. Budding
D. Conjugation

✓ **Correct Answer: B. Asexual reproduction**

Explanation: Binary fission produces daughter cells from one parent cell without fusion of gametes, so it is asexual reproduction.

18. Conjugation in Paramecium results primarily in:

A. Immediate increase in population
B. Genetic recombination
C. Formation of pseudopodia
D. Formation of spores

✓ **Correct Answer: B. Genetic recombination**

Explanation: Conjugation exchanges genetic material between cells, increasing genetic variation rather than directly increasing population size.

CHAPTER TWO — PORIFERA, CNIDARIA & CTENOPHORA

19. The phylum Porifera includes:

A. Jellyfish
B. Sponges
C. Flatworms
D. Roundworms

✓ **Correct Answer: B. Sponges**

Explanation: Porifera is the animal phylum containing sponges.

20. The characteristic feature of Porifera is the presence of:

A. Jointed appendages
B. Numerous pores
C. Notochord
D. Segmentation

✓ **Correct Answer: B. Numerous pores**

Explanation: The name Porifera means pore-bearing, referring to the many openings through which water enters the body.

21. Water enters a sponge mainly through:

A. Osculum
B. Ostia
C. Mouth
D. Anus

✓ **Correct Answer: B. Ostia**

Explanation: Ostia are small pores through which water enters the sponge's canal system.

22. Water leaves a sponge through the:

A. Ostia
B. Spongocoel
C. Osculum
D. Mesoglea

✓ **Correct Answer: C. Osculum**

Explanation: The osculum is the larger opening through which water exits the sponge.

23. The cells that create water currents in sponges are:

A. Cnidocytes
B. Choanocytes
C. Nephridia
D. Flame cells

✓ **Correct Answer: B. Choanocytes**

Explanation: Choanocytes have beating flagella that drive water currents and collars that trap food particles.

24. The skeleton of many sponges consists of:

A. Chitin only
B. Spicules and/or spongin
C. Keratin
D. Bone

✓ **Correct Answer: B. Spicules and/or spongin**

Explanation: Depending on the sponge, the supporting skeleton may contain mineral spicules, spongin fibers, or both.

25. Cnidarians are characterized by:

A. Cnidocytes
B. Nephridia
C. Jointed legs
D. Notochord

✓ **Correct Answer: A. Cnidocytes**

Explanation: Cnidocytes are specialized stinging cells unique to cnidarians.

26. The stinging organelle within a cnidocyte is the:

A. Nematocyst
B. Choanocyte
C. Statocyst
D. Colloblast

✓ **Correct Answer: A. Nematocyst**

Explanation: A nematocyst is the specialized capsule within a cnidocyte that can discharge a thread for prey capture or defense.

27. The body symmetry of most cnidarians is:

A. Bilateral
B. Radial
C. Asymmetrical
D. Spherical only

✓ **Correct Answer: B. Radial**

Explanation: Radial symmetry allows a cnidarian to interact with the environment from several directions.

28. Which of the following has a medusa stage?

A. Jellyfish
B. Earthworm
C. Sponge
D. Tapeworm

✓ **Correct Answer: A. Jellyfish**

Explanation: Jellyfish are the free-swimming medusa form of cnidarians.

29. The sessile form of a cnidarian is called:

A. Medusa
B. Polyp
C. Larva
D. Trophozoite

✓ **Correct Answer: B. Polyp**

Explanation: A polyp is generally attached to a substrate, whereas a medusa is usually free-swimming.

30. Twisted: Which statement correctly distinguishes a polyp from a medusa?

A. Polyp is usually free-swimming while medusa is attached
B. Polyp is usually attached while medusa is usually free-swimming
C. Both are always attached
D. Both are always free-swimming

✓ **Correct Answer: B. Polyp is usually attached while medusa is usually free-swimming**

Explanation: The typical polyp is sessile and cylindrical, while the medusa is usually free-swimming and umbrella-shaped.

31. Ctenophores are commonly called:

A. Comb jellies
B. Sea sponges
C. Flatworms
D. Sea anemones

✓ **Correct Answer: A. Comb jellies**

Explanation: Ctenophores are popularly called comb jellies because of their rows of ciliary comb plates.

32. Locomotion in ctenophores is mainly by:

A. Flagella
B. Cilia arranged in comb rows
C. Pseudopodia
D. Parapodia

✓ **Correct Answer: B. Cilia arranged in comb rows**

Explanation: Eight rows of fused cilia called comb plates beat rhythmically and propel the animal.

33. The adhesive cells of ctenophores are called:

A. Cnidocytes
B. Colloblasts
C. Choanocytes
D. Flame cells

✓ **Correct Answer: B. Colloblasts**

Explanation: Colloblasts release sticky material that helps ctenophores capture prey.

34. Which group lacks true tissues?

A. Cnidaria
B. Ctenophora
C. Porifera
D. Annelida

✓ **Correct Answer: C. Porifera**

Explanation: Sponges have cellular-level organization and lack true tissues.

CHAPTER THREE — COELOMATE ANIMALS

35. A coelom is:

A. A digestive cavity
B. A body cavity completely lined by mesoderm
C. A respiratory organ
D. A reproductive opening

✓ **Correct Answer: B. A body cavity completely lined by mesoderm**

Explanation: A true coelom is a fluid-filled body cavity completely surrounded by mesoderm.

36. Animals without a body cavity are described as:

A. Coelomates
B. Pseudocoelomates
C. Acoelomates
D. Deuterostomes

✓ **Correct Answer: C. Acoelomates**

Explanation: Acoelomates lack a body cavity between the gut and body wall.

37. Animals whose body cavity is not completely lined by mesoderm are:

A. Acoelomates
B. Pseudocoelomates
C. Coelomates
D. Diploblasts

✓ **Correct Answer: B. Pseudocoelomates**

Explanation: A pseudocoel is not completely enclosed by mesoderm, distinguishing pseudocoelomates from true coelomates.

38. Which of the following is a true coelomate?

A. Earthworm
B. Tapeworm
C. Roundworm
D. Planarian

✓ **Correct Answer: A. Earthworm**

Explanation: Earthworms are annelids and possess a true coelom.

39. The coelom can serve as:

A. A hydrostatic skeleton
B. A digestive enzyme
C. A respiratory pigment
D. A sensory organ

✓ **Correct Answer: A. A hydrostatic skeleton**

Explanation: Fluid within the coelom can provide support and assist movement as a hydrostatic skeleton.

40. Animals with two embryonic germ layers are:

A. Triploblastic
B. Diploblastic
C. Coelomate
D. Segmented

✓ **Correct Answer: B. Diploblastic**

Explanation: Diploblastic animals develop from two primary germ layers: ectoderm and endoderm.

41. The three primary germ layers are:

A. Ectoderm, mesoderm and endoderm
B. Epidermis, dermis and mesoderm
C. Ectoderm, epidermis and endoderm
D. Mesoderm, coelom and ectoderm

✓ **Correct Answer: A. Ectoderm, mesoderm and endoderm**

Explanation: Triploblastic animals develop ectoderm, mesoderm and endoderm.

42. Animals with bilateral symmetry generally have:

A. Only one body axis
B. Distinct anterior and posterior ends
C. No nervous system
D. No mesoderm

✓ **Correct Answer: B. Distinct anterior and posterior ends**

Explanation: Bilateral symmetry is associated with directional movement and differentiation of anterior and posterior ends.

CHAPTER FOUR — PHYLUM ANNELIDA

43. Annelids are commonly characterized by:

A. Segmentation
B. Radial symmetry
C. Jointed appendages
D. A water vascular system

✓ **Correct Answer: A. Segmentation**

Explanation: Annelids have segmented bodies, a feature called metamerism.

44. The earthworm belongs to:

A. Mollusca
B. Annelida
C. Arthropoda
D. Nematoda

✓ **Correct Answer: B. Annelida**

Explanation: Earthworms are segmented worms in the phylum Annelida.

45. The body cavity of annelids is:

A. Absent
B. A pseudocoelom
C. A true coelom
D. A haemocoel only

✓ **Correct Answer: C. A true coelom**

Explanation: Annelids are true coelomates; their coelom is lined by mesoderm.

46. The external division of an annelid body into segments is called:

A. Metamerism
B. Cephalization
C. Tagmatization
D. Budding

✓ **Correct Answer: A. Metamerism**

Explanation: Metamerism refers to serial repetition of body segments.

47. Earthworms belong to the class:

A. Polychaeta
B. Oligochaeta
C. Hirudinea
D. Turbellaria

✓ **Correct Answer: B. Oligochaeta**

Explanation: Earthworms are traditionally placed in Oligochaeta, characterized by relatively few setae.

48. Setae in earthworms are mainly useful for:

A. Digestion
B. Locomotion
C. Respiration
D. Excretion

✓ **Correct Answer: B. Locomotion**

Explanation: Setae grip the soil and provide traction as the circular and longitudinal muscles contract.

49. The clitellum of an earthworm is important in:

A. Digestion
B. Reproduction
C. Respiration
D. Circulation

✓ **Correct Answer: B. Reproduction**

Explanation: The clitellum secretes material that forms the cocoon and is important during reproduction.

50. Earthworms respire mainly through:

A. Gills
B. Tracheae
C. Moist skin
D. Lungs

✓ **Correct Answer: C. Moist skin**

Explanation: Oxygen and carbon dioxide diffuse through the moist body wall; drying interferes with this exchange.

51. Twisted: If the skin of an earthworm becomes excessively dry, the most immediate physiological problem is likely to be:

A. Reduced digestion
B. Impaired gaseous exchange
C. Failure of segmentation
D. Loss of setae

✓ **Correct Answer: B. Impaired gaseous exchange**

Explanation: Moisture is essential for diffusion of respiratory gases across the earthworm's skin.

52. Leeches belong to:

A. Hirudinea
B. Oligochaeta
C. Polychaeta
D. Gastropoda

✓ **Correct Answer: A. Hirudinea**

Explanation: Leeches are traditionally classified in Hirudinea.

53. Polychaetes are characterized by:

A. Numerous parapodia and setae
B. Absence of segmentation
C. Jointed legs
D. A mantle

✓ **Correct Answer: A. Numerous parapodia and setae**

Explanation: Polychaetes are segmented annelids with many setae, commonly carried on parapodia.

54. The circulatory system of annelids is generally:

A. Open
B. Closed
C. Absent
D. Haemocoelic

✓ **Correct Answer: B. Closed**

Explanation: Blood remains within vessels in the typical annelid closed circulatory system.

CHAPTER FIVE — PHYLUM MOLLUSCA

55. The largest class of Mollusca is:

A. Bivalvia
B. Gastropoda
C. Cephalopoda
D. Polyplacophora

✓ **Correct Answer: B. Gastropoda**

Explanation: Gastropoda is the largest molluscan class and includes snails and slugs.

56. Which animal belongs to Mollusca?

A. Snail
B. Earthworm
C. Grasshopper
D. Starfish

✓ **Correct Answer: A. Snail**

Explanation: Snails are gastropod molluscs.

57. The body of a typical mollusc consists of:

A. Head, thorax and abdomen
B. Head, muscular foot and visceral mass
C. Cephalothorax and abdomen
D. Proboscis and trunk

✓ **Correct Answer: B. Head, muscular foot and visceral mass**

Explanation: The basic molluscan body plan includes a muscular foot, visceral mass and mantle; a distinct head is present in many but not all.

58. The mantle of a mollusc primarily:

A. Produces the shell in shelled forms
B. Pumps blood
C. Produces digestive enzymes only
D. Forms the nervous system

✓ **Correct Answer: A. Produces the shell in shelled forms**

Explanation: The mantle is a body-wall fold that secretes the shell in shelled molluscs and also contributes to the mantle cavity.

59. The radula is primarily used for:

A. Locomotion
B. Scraping/feeding
C. Respiration
D. Excretion

✓ **Correct Answer: B. Scraping/feeding**

Explanation: The radula is a toothed feeding structure used by most molluscs to scrape or cut food.

60. Which molluscs generally lack a radula?

A. Gastropods
B. Cephalopods
C. Bivalves
D. Chitons

✓ **Correct Answer: C. Bivalves**

Explanation: Bivalves are filter feeders and do not need a radula for scraping food.

61. Which is a cephalopod?

A. Snail
B. Oyster
C. Octopus
D. Mussel

✓ **Correct Answer: C. Octopus**

Explanation: Octopus is a cephalopod, along with squid, cuttlefish and nautilus.

62. Cephalopods are generally:

A. Poorly developed molluscs
B. Highly active predators
C. Sessile filter feeders
D. Parasites

✓ **Correct Answer: B. Highly active predators**

Explanation: Cephalopods have advanced nervous systems, eyes, locomotion and predatory adaptations.

63. Bivalves obtain food mainly through:

A. Predation
B. Filter feeding
C. Scraping algae with radula
D. Absorption through the skin

✓ **Correct Answer: B. Filter feeding**

Explanation: Bivalves use their gills to filter suspended particles from water.

64. Twisted: A mollusc with two shells hinged together and no distinct head is most likely a:

A. Gastropod
B. Bivalve
C. Cephalopod
D. Chiton

✓ **Correct Answer: B. Bivalve**

Explanation: Bivalves have two hinged valves and generally lack a distinct head and radula.

65. Torsion is particularly characteristic of:

A. Gastropods
B. Bivalves
C. Cephalopods
D. Annelids

✓ **Correct Answer: A. Gastropods**

Explanation: Torsion is a developmental rotation of the visceral mass and mantle cavity characteristic of gastropods.

66. The foot of molluscs is mainly associated with:

A. Locomotion
B. Digestion
C. Circulation
D. Reproduction only

✓ **Correct Answer: A. Locomotion**

Explanation: The muscular foot is commonly used for crawling, burrowing or other forms of locomotion, although it is modified in different molluscs.

CHAPTER SIX — PHYLUM ARTHROPODA

67. The largest animal phylum is:

A. Mollusca
B. Arthropoda
C. Annelida
D. Chordata

✓ **Correct Answer: B. Arthropoda**

Explanation: Arthropoda contains the greatest number of described animal species.

68. Arthropods are characterized by:

A. Jointed appendages
B. A notochord
C. A mantle
D. Radial symmetry

✓ **Correct Answer: A. Jointed appendages**

Explanation: Jointed appendages are a defining arthropod feature.

69. The exoskeleton of arthropods is mainly composed of:

A. Cellulose
B. Chitin
C. Keratin
D. Calcium phosphate only

✓ **Correct Answer: B. Chitin**

Explanation: The arthropod exoskeleton is primarily chitinous, often reinforced with proteins and minerals.

70. Growth in arthropods requires:

A. Budding
B. Molting
C. Conjugation
D. Fragmentation

✓ **Correct Answer: B. Molting**

Explanation: The rigid exoskeleton cannot expand continuously, so arthropods shed it during growth.

71. The shedding of an arthropod's old exoskeleton is called:

A. Metamorphosis
B. Ecdysis
C. Torsion
D. Regeneration

✓ **Correct Answer: B. Ecdysis**

Explanation: Ecdysis is the process of shedding the old cuticle or exoskeleton.

72. Insects possess:

A. Two pairs of legs
B. Three pairs of legs
C. Four pairs of legs
D. Five pairs of legs

✓ **Correct Answer: B. Three pairs of legs**

Explanation: Adult insects have three pairs of thoracic legs.

73. Arachnids possess:

A. Three pairs of legs
B. Four pairs of legs
C. Five pairs of legs
D. Six pairs of legs

✓ **Correct Answer: B. Four pairs of legs**

Explanation: Adult arachnids typically have four pairs of walking legs.

74. Which of the following is an arachnid?

A. Cockroach
B. Crab
C. Spider
D. Centipede

✓ **Correct Answer: C. Spider**

Explanation: Spiders are arachnids, a major arthropod group.

75. Crustaceans generally possess:

A. One pair of antennae
B. Two pairs of antennae
C. No antennae
D. Three pairs of antennae

✓ **Correct Answer: B. Two pairs of antennae**

Explanation: Two pairs of antennae are a characteristic feature of many crustaceans.

76. Insects possess:

A. Two pairs of antennae
B. One pair of antennae
C. No antennae
D. Four pairs of antennae

✓ **Correct Answer: B. One pair of antennae**

Explanation: Insects have one pair of antennae attached to the head.

77. The body of an insect is divided into:

A. Head and abdomen
B. Cephalothorax and abdomen
C. Head, thorax and abdomen
D. Head, trunk and tail

✓ **Correct Answer: C. Head, thorax and abdomen**

Explanation: The insect body has three tagmata: head, thorax and abdomen.

78. The body of a spider is divided into:

A. Head, thorax and abdomen
B. Cephalothorax and abdomen
C. Head and abdomen only
D. Thorax and abdomen

✓ **Correct Answer: B. Cephalothorax and abdomen**

Explanation: Arachnids such as spiders have a cephalothorax (prosoma) and abdomen (opisthosoma).

79. The respiratory organs of insects are:

A. Gills
B. Lungs
C. Tracheae
D. Nephridia

✓ **Correct Answer: C. Tracheae**

Explanation: Insects exchange gases through a network of tracheae and tracheoles connected to spiracles.

80. Twisted: Which combination is correctly matched?

A. Insect — four pairs of legs
B. Arachnid — three pairs of legs
C. Crustacean — two pairs of antennae
D. Insect — two pairs of antennae

✓ **Correct Answer: C. Crustacean — two pairs of antennae**

Explanation: Two pairs of antennae are characteristic of crustaceans; insects have one pair and three pairs of legs.

81. Complete metamorphosis involves:

A. Egg → nymph → adult
B. Egg → larva → pupa → adult
C. Egg → adult only
D. Larva → nymph → adult

✓ **Correct Answer: B. Egg → larva → pupa → adult**

Explanation: Holometabolous development has distinct egg, larval, pupal and adult stages.

82. Which insect undergoes complete metamorphosis?

A. Grasshopper
B. Cockroach
C. Butterfly
D. Termite

✓ **Correct Answer: C. Butterfly**

Explanation: Butterflies develop through egg, larva (caterpillar), pupa and adult stages.

CHAPTER SEVEN — PLATYHELMINTHES, ROTIFERA, NEMATODA & ACANTHOCEPHALA

83. Platyhelminthes are commonly called:

A. Roundworms
B. Flatworms
C. Segmented worms
D. Hookworms

✓ **Correct Answer: B. Flatworms**

Explanation: Platyhelminthes are dorsoventrally flattened worms.

84. Platyhelminths are:

A. Coelomates
B. Pseudocoelomates
C. Acoelomates
D. Deuterostomes

✓ **Correct Answer: C. Acoelomates**

Explanation: Platyhelminths lack a fluid-filled body cavity between the gut and body wall.

85. Which of the following is a flatworm?

A. Taenia
B. Ascaris
C. Earthworm
D. Leech

✓ **Correct Answer: A. Taenia**

Explanation: Taenia is a tapeworm belonging to Platyhelminthes.

86. The body of a tapeworm is divided into:

A. Segments called proglottids
B. Metameres with parapodia
C. Thoracic segments
D. Tagmata

✓ **Correct Answer: A. Segments called proglottids**

Explanation: Tapeworms consist of a scolex, neck and a chain of proglottids.

87. The scolex of a tapeworm is mainly used for:

A. Digestion
B. Attachment
C. Respiration
D. Reproduction only

✓ **Correct Answer: B. Attachment**

Explanation: The scolex bears structures such as suckers and, in some species, hooks for attachment to the host.

88. Tapeworms lack a:

A. Nervous system
B. Digestive system
C. Reproductive system
D. Excretory system

✓ **Correct Answer: B. Digestive system**

Explanation: Tapeworms have no digestive tract; they absorb nutrients through their tegument.

89. Tapeworms obtain nutrients mainly by:

A. Ingesting food through a mouth
B. Absorbing nutrients across their body surface
C. Photosynthesis
D. Filtering water

✓ **Correct Answer: B. Absorbing nutrients across their body surface**

Explanation: Their tegument is specialized for absorbing dissolved nutrients from the host intestine.

90. Schistosoma causes:

A. Malaria
B. Schistosomiasis
C. Filariasis
D. Sleeping sickness

✓ **Correct Answer: B. Schistosomiasis**

Explanation: Schistosoma species are parasitic blood flukes that cause schistosomiasis.

91. The intermediate host of Schistosoma is usually a:

A. Mosquito
B. Snail
C. Fish
D. Crab

✓ **Correct Answer: B. Snail**

Explanation: Freshwater snails act as intermediate hosts for the development of Schistosoma larvae.

92. Nematodes are commonly known as:

A. Flatworms
B. Roundworms
C. Segmented worms
D. Ribbon worms

✓ **Correct Answer: B. Roundworms**

Explanation: Nematodes are cylindrical, unsegmented roundworms.

93. Ascaris lumbricoides is a:

A. Flatworm
B. Roundworm
C. Annelid
D. Mollusc

✓ **Correct Answer: B. Roundworm**

Explanation: Ascaris is a parasitic nematode of humans.

94. Nematodes generally have:

A. No body cavity
B. A true coelom
C. A pseudocoelom
D. A haemocoel

✓ **Correct Answer: C. A pseudocoelom**

Explanation: The pseudocoel is a body cavity not completely lined by mesoderm.

95. Twisted: Which organism is most likely to have a complete digestive tract with both mouth and anus?

A. Tapeworm
B. Planarian
C. Nematode
D. Sponge

✓ **Correct Answer: C. Nematode**

Explanation: Nematodes have a complete digestive tract with separate mouth and anus; tapeworms lack a digestive tract.

96. Rotifers are characterized by the presence of:

A. Jointed legs
B. Corona
C. Radula
D. Notochord

✓ **Correct Answer: B. Corona**

Explanation: The corona is a ciliated, wheel-like structure characteristic of rotifers.

97. The corona of a rotifer is associated mainly with:

A. Locomotion and feeding
B. Reproduction only
C. Excretion only
D. Circulation

✓ **Correct Answer: A. Locomotion and feeding**

Explanation: Beating cilia of the corona create water currents for feeding and can also aid movement.

98. Acanthocephalans are commonly called:

A. Thorny-headed worms
B. Flatworms
C. Roundworms
D. Ribbon worms

✓ **Correct Answer: A. Thorny-headed worms**

Explanation: Acanthocephalans have a characteristic hooked or spiny proboscis, hence the name thorny-headed worms.

99. Acanthocephalans possess a characteristic:

A. Radula
B. Eversible spiny proboscis
C. Water vascular system
D. Notochord

✓ **Correct Answer: B. Eversible spiny proboscis**

Explanation: The hooked proboscis anchors the parasite to the host's intestinal wall.

CHAPTER EIGHT — CHORDATES

100. The defining feature of chordates is the presence of a:

A. Mantle
B. Notochord
C. Radula
D. Exoskeleton

✓ **Correct Answer: B. Notochord**

Explanation: The notochord is one of the four fundamental chordate characteristics.

101. Which of the following is NOT a fundamental chordate characteristic?

A. Notochord
B. Dorsal hollow nerve cord
C. Pharyngeal slits
D. Ventral solid nerve cord

✓ **Correct Answer: D. Ventral solid nerve cord**

Explanation: Chordates have a dorsal, hollow nerve cord rather than a ventral solid nerve cord.

102. The nerve cord of chordates is:

A. Ventral and solid
B. Dorsal and hollow
C. Lateral and hollow
D. Ventral and hollow

✓ **Correct Answer: B. Dorsal and hollow**

Explanation: A dorsal hollow nerve cord is a defining chordate feature.

103. The notochord is located:

A. Dorsal to the nerve cord
B. Ventral to the nerve cord
C. Lateral to the gut
D. Inside the gut

✓ **Correct Answer: B. Ventral to the nerve cord**

Explanation: In the chordate body plan, the notochord lies below the dorsal nerve cord.

104. Which structure develops into the vertebral column in vertebrates?

A. Notochord
B. Neural tube
C. Archenteron
D. Coelom

✓ **Correct Answer: A. Notochord**

Explanation: In vertebrate development, the notochord provides axial support and is largely replaced by the vertebral column, although remnants persist in intervertebral discs.

105. Chordates are generally:

A. Radially symmetrical
B. Bilaterally symmetrical
C. Asymmetrical
D. Spherically symmetrical

✓ **Correct Answer: B. Bilaterally symmetrical**

Explanation: Chordates have bilateral symmetry, with distinct left and right sides.

106. The three major chordate groups traditionally include:

A. Urochordata, Cephalochordata and Vertebrata
B. Porifera, Cnidaria and Mollusca
C. Pisces, Amphibia and Reptilia
D. Protozoa, Rotifera and Nematoda

✓ **Correct Answer: A. Urochordata, Cephalochordata and Vertebrata**

Explanation: These are the traditional major divisions used to describe chordate diversity.

107. In urochordates, the notochord is mainly present in the:

A. Adult head
B. Larval tail
C. Adult trunk
D. Entire adult body

✓ **Correct Answer: B. Larval tail**

Explanation: In tunicates, the notochord is prominent in the larval tail and is lost during adult metamorphosis.

108. Twisted: Which chordate characteristic may be present only during embryonic or larval development in some chordates?

A. Notochord
B. Mouth
C. Digestive tract
D. Circulatory system

✓ **Correct Answer: A. Notochord**

Explanation: In many vertebrates and tunicates, the notochord is prominent early in development but becomes modified or replaced later.

CHAPTER NINE — PISCES & AMPHIBIA

109. Fish primarily respire using:

A. Lungs
B. Gills
C. Tracheae
D. Skin only

✓ **Correct Answer: B. Gills**

Explanation: Gills provide a large, thin, vascular surface for extracting dissolved oxygen from water.

110. The body of most fishes is covered with:

A. Feathers
B. Hair
C. Scales
D. Fur

✓ **Correct Answer: C. Scales**

Explanation: Most fish have scales covered by mucus, although there are exceptions.

111. The organ responsible for maintaining buoyancy in many bony fishes is the:

A. Swim bladder
B. Lateral line
C. Operculum
D. Cloaca

✓ **Correct Answer: A. Swim bladder**

Explanation: The swim bladder is a gas-filled organ that helps many bony fishes control buoyancy.

112. The lateral line system of fish detects:

A. Blood pressure
B. Water movement and vibrations
C. Light intensity only
D. Air pressure only

✓ **Correct Answer: B. Water movement and vibrations**

Explanation: Sensory receptors of the lateral line detect movements and pressure changes in surrounding water.

113. The operculum in bony fish:

A. Covers the gills
B. Produces eggs
C. Stores food
D. Controls buoyancy

✓ **Correct Answer: A. Covers the gills**

Explanation: The operculum is a bony flap covering and protecting the gills and helps pump water across them.

114. Fish are generally:

A. Endothermic
B. Ectothermic
C. Homeothermic
D. Warm-blooded mammals

✓ **Correct Answer: B. Ectothermic**

Explanation: Most fishes rely heavily on environmental heat to determine body temperature, so they are ectothermic.

115. Amphibians generally spend:

A. Their entire lives in water
B. Their entire lives on land
C. Part of their life in water and part on land
D. Their entire lives underground

✓ **Correct Answer: C. Part of their life in water and part on land**

Explanation: Many amphibians have aquatic eggs/larvae and more terrestrial adult stages.

116. The larval stage of a frog is called:

A. Nymph
B. Tadpole
C. Pupa
D. Planula

✓ **Correct Answer: B. Tadpole**

Explanation: A frog larva is called a tadpole and typically has a tail and gills.

117. Tadpoles generally respire using:

A. Lungs
B. Gills
C. Tracheae
D. Spiracles

✓ **Correct Answer: B. Gills**

Explanation: Tadpoles are aquatic and use gills, especially early in development.

118. Adult frogs respire through:

A. Lungs and skin
B. Gills only
C. Tracheae only
D. Book lungs

✓ **Correct Answer: A. Lungs and skin**

Explanation: Adult frogs use lungs and also exchange gases through their moist skin.

119. Twisted: The ability of an adult frog to exchange gases through its skin is most effective when the skin is:

A. Dry
B. Moist
C. Covered with scales
D. Thick and keratinized

✓ **Correct Answer: B. Moist**

Explanation: A moist surface allows oxygen and carbon dioxide to dissolve and diffuse through the skin.

120. Metamorphosis in amphibians involves transformation from:

A. Egg to pupa
B. Tadpole to adult
C. Larva to nymph
D. Egg directly to adult

✓ **Correct Answer: B. Tadpole to adult**

Explanation: Frog metamorphosis changes an aquatic tadpole into the adult frog form.

CHAPTER TEN — REPTILIA & AVES

121. Reptiles are generally characterized by:

A. Moist glandular skin
B. Dry, keratinized skin
C. Feathers
D. Hair

✓ **Correct Answer: B. Dry, keratinized skin**

Explanation: Keratinized skin and scales reduce water loss and help reptiles live successfully on land.

122. The main respiratory organs of reptiles are:

A. Gills
B. Lungs
C. Skin
D. Tracheae

✓ **Correct Answer: B. Lungs**

Explanation: Reptiles use lungs for gas exchange and generally do not rely on skin respiration.

123. Reptiles are generally:

A. Endothermic
B. Ectothermic
C. Homeothermic
D. Warm-blooded

✓ **Correct Answer: B. Ectothermic**

Explanation: Most reptiles depend significantly on external heat sources for thermoregulation.

124. The amniotic egg is an important adaptation for:

A. Aquatic reproduction only
B. Reproduction away from water
C. Respiration through gills
D. Filter feeding

✓ **Correct Answer: B. Reproduction away from water**

Explanation: Extraembryonic membranes allow the embryo to develop in a protected fluid environment on land.

125. Which of the following is NOT a reptile?

A. Crocodile
B. Turtle
C. Lizard
D. Frog

✓ **Correct Answer: D. Frog**

Explanation: A frog is an amphibian, not a reptile.

126. Birds are characterized by:

A. Scales only
B. Feathers
C. Hair
D. Moist skin

✓ **Correct Answer: B. Feathers**

Explanation: Feathers are a defining characteristic of birds and support insulation, display and flight.

127. The forelimbs of birds are modified into:

A. Fins
B. Wings
C. Flippers
D. Paddles only

✓ **Correct Answer: B. Wings**

Explanation: Bird forelimbs are modified into wings, although wing shape and flight ability vary among species.

128. The respiratory system of birds is highly efficient partly because of:

A. Air sacs
B. Gills
C. Skin respiration
D. Book lungs

✓ **Correct Answer: A. Air sacs**

Explanation: Air sacs help maintain one-way airflow through the lungs and support continuous gas exchange.

129. Birds are:

A. Ectothermic
B. Endothermic
C. Poikilothermic only
D. Cold-blooded

✓ **Correct Answer: B. Endothermic**

Explanation: Birds generate metabolic heat and maintain a relatively stable internal body temperature.

130. The bones of many flying birds are:

A. Extremely solid and heavy
B. Pneumatic/lightened
C. Made entirely of cartilage
D. Completely absent

✓ **Correct Answer: B. Pneumatic/lightened**

Explanation: Many birds have hollow or pneumatic bones, reducing mass while maintaining structural strength.

131. The covering of a bird's body is primarily:

A. Hair
B. Feathers
C. Scales only
D. Mucus

✓ **Correct Answer: B. Feathers**

Explanation: Feathers form the characteristic external covering of birds.

132. Birds possess a:

A. Three-chambered heart
B. Four-chambered heart
C. Two-chambered heart
D. One-chambered heart

✓ **Correct Answer: B. Four-chambered heart**

Explanation: A four-chambered heart completely separates pulmonary and systemic blood flow.

133. Twisted: Which combination is correct?

A. Reptiles — feathers — endothermic
B. Birds — feathers — endothermic
C. Amphibians — hair — endothermic
D. Fish — feathers — endothermic

✓ **Correct Answer: B. Birds — feathers — endothermic**

Explanation: Birds have feathers and are endothermic; the other combinations contain incorrect characteristics.

CHAPTER ELEVEN — MAMMALIA

134. The defining characteristic of mammals is the presence of:

A. Feathers
B. Mammary glands
C. Gills
D. Scales

✓ **Correct Answer: B. Mammary glands**

Explanation: Mammary glands produce milk used to nourish young, a key mammalian characteristic.

135. Mammals are generally:

A. Ectothermic
B. Endothermic
C. Poikilothermic
D. Cold-blooded

✓ **Correct Answer: B. Endothermic**

Explanation: Mammals generate substantial metabolic heat and regulate body temperature internally.

136. Mammalian skin typically possesses:

A. Hair
B. Feathers
C. Scales only
D. Cilia

✓ **Correct Answer: A. Hair**

Explanation: Hair is a characteristic mammalian structure involved in insulation, sensation and protection.

137. The mammalian heart has:

A. Two chambers
B. Three chambers
C. Four chambers
D. Five chambers

✓ **Correct Answer: C. Four chambers**

Explanation: Two atria and two ventricles completely separate oxygenated from deoxygenated blood.

138. The diaphragm in mammals is primarily involved in:

A. Digestion
B. Respiration
C. Excretion
D. Reproduction

✓ **Correct Answer: B. Respiration**

Explanation: Contraction of the diaphragm increases thoracic volume and draws air into the lungs.

139. Mammals generally possess:

A. Three middle-ear bones
B. One middle-ear bone
C. No middle-ear bones
D. Five middle-ear bones

✓ **Correct Answer: A. Three middle-ear bones**

Explanation: The malleus, incus and stapes transmit sound vibrations from the eardrum to the inner ear.

140. Which of the following is a monotreme?

A. Human
B. Platypus
C. Dog
D. Kangaroo

✓ **Correct Answer: B. Platypus**

Explanation: The platypus is a monotreme, one of the few living mammals that lay eggs.

141. Monotremes are unusual mammals because they:

A. Have feathers
B. Lay eggs
C. Lack mammary glands
D. Have gills

✓ **Correct Answer: B. Lay eggs**

Explanation: Monotremes lay eggs but still possess mammalian features such as hair and mammary glands.

142. Marsupials give birth to:

A. Fully developed young only
B. Relatively undeveloped young
C. Eggs
D. Larvae

✓ **Correct Answer: B. Relatively undeveloped young**

Explanation: Marsupial young are born at an early developmental stage and continue development after birth.

143. The young of most marsupials continue development in the:

A. Cloaca
B. Pouch
C. Uterus only
D. Egg shell

✓ **Correct Answer: B. Pouch**

Explanation: The pouch provides a protected environment where the young attaches to a teat and continues development.

144. Placental mammals are characterized by:

A. Development of the embryo in a uterus with a placenta
B. Egg laying
C. Gills
D. External fertilization

✓ **Correct Answer: A. Development of the embryo in a uterus with a placenta**

Explanation: In placental mammals, the placenta supports exchange of gases, nutrients and wastes between mother and fetus.

145. Humans belong to the order:

A. Rodentia
B. Primates
C. Carnivora
D. Artiodactyla

✓ **Correct Answer: B. Primates**

Explanation: Humans are primates, a mammalian order that includes apes, monkeys and related groups.

146. Which of the following is NOT a mammal?

A. Whale
B. Bat
C. Dolphin
D. Shark

✓ **Correct Answer: D. Shark**

Explanation: Sharks are cartilaginous fishes. Whales, bats and dolphins are mammals.

147. Twisted: A bat is classified as a mammal rather than a bird mainly because it:

A. Can fly
B. Has wings
C. Possesses mammary glands and hair
D. Lays eggs

✓ **Correct Answer: C. Possesses mammary glands and hair**

Explanation: Flight and wings occur in both bats and birds, but mammary glands and hair are mammalian characteristics.

148. Whales are mammals because they:

A. Live in water
B. Have fins
C. Breathe with lungs and possess mammalian characteristics
D. Have gills

✓ **Correct Answer: C. Breathe with lungs and possess mammalian characteristics**

Explanation: Whales are aquatic mammals that breathe air with lungs and have other mammalian features.

149. The teeth of mammals are generally:

A. All identical
B. Differentiated into different types
C. Absent
D. Made of cartilage

✓ **Correct Answer: B. Differentiated into different types**

Explanation: Mammalian teeth are typically heterodont, with incisors, canines, premolars and molars specialized for different functions.

150. Twisted: Which statement is TRUE?

A. All mammals lay eggs
B. All mammals live on land
C. All mammals possess mammary glands
D. All mammals have pouches

✓ **Correct Answer: C. All mammals possess mammary glands**

Explanation: Mammary glands are a defining mammalian feature, while egg laying, terrestrial life and pouches are not universal.

□ HIGH-YIELD MIXED/TWISTED QUESTIONS

151. Which group contains animals that are predominantly unicellular?

A. Porifera
B. Protozoa
C. Cnidaria
D. Annelida

✓ **Correct Answer: B. Protozoa**

Explanation: Protozoans are predominantly unicellular, whereas the other listed groups are multicellular animals.

152. Which of the following has the simplest body organization?

A. Mammal
B. Annelid
C. Sponge
D. Arthropod

✓ **Correct Answer: C. Sponge**

Explanation: Sponges have cellular-level organization and lack the true tissues and organs seen in more complex animals.

153. Which pair is INCORRECTLY matched?

A. Porifera — choanocytes
B. Cnidaria — cnidocytes
C. Ctenophora — colloblasts
D. Annelida — radula

✓ **Correct Answer: D. Annelida — radula**

Explanation: Radula is a molluscan feeding structure, not an annelid structure.

154. Which animal is an acoelomate?

A. Earthworm
B. Tapeworm
C. Roundworm
D. Mollusc

✓ **Correct Answer: B. Tapeworm**

Explanation: Tapeworms are platyhelminths and lack a body cavity, making them acoelomates.

155. Which animal is a pseudocoelomate?

A. Ascaris
B. Taenia
C. Earthworm
D. Snail

✓ **Correct Answer: A. Ascaris**

Explanation: Ascaris is a nematode, and nematodes possess a pseudocoelom.

156. Which group possesses a true coelom?

A. Platyhelminthes
B. Nematoda
C. Annelida
D. Rotifera

✓ **Correct Answer: C. Annelida**

Explanation: Annelids possess a true mesoderm-lined coelom.

157. Which organism possesses jointed appendages?

A. Earthworm
B. Spider
C. Snail
D. Tapeworm

✓ **Correct Answer: B. Spider**

Explanation: Spiders are arthropods, and jointed appendages are a defining arthropod feature.

158. Which organism possesses a radula?

A. Bivalve
B. Gastropod
C. Tapeworm
D. Earthworm

✓ **Correct Answer: B. Gastropod**

Explanation: Gastropods generally possess a radula for feeding; bivalves do not.

159. Which organism lacks a digestive tract?

A. Earthworm
B. Tapeworm
C. Nematode
D. Snail

✓ **Correct Answer: B. Tapeworm**

Explanation: Tapeworms absorb nutrients through their tegument and therefore do not have a digestive tract.

160. Which animal possesses a dorsal hollow nerve cord?

A. Earthworm
B. Grasshopper
C. Frog
D. Snail

✓ **Correct Answer: C. Frog**

Explanation: Frogs are chordates and therefore possess the chordate dorsal hollow nerve cord at least during development.

161. Which group is characterized by feathers?

A. Reptilia
B. Aves
C. Mammalia
D. Amphibia

✓ **Correct Answer: B. Aves**

Explanation: Feathers are the defining external structures of birds.

162. Which group is characterized by hair and mammary glands?

A. Aves
B. Reptilia
C. Mammalia
D. Amphibia

✓ **Correct Answer: C. Mammalia**

Explanation: Hair and mammary glands are characteristic mammalian features.

163. Which of the following is ectothermic?

A. Bird
B. Mammal
C. Frog
D. Human

✓ **Correct Answer: C. Frog**

Explanation: Frogs are ectothermic; birds and mammals are endothermic.

164. Which of the following is endothermic?

A. Lizard
B. Frog
C. Fish
D. Bird

✓ **Correct Answer: D. Bird**

Explanation: Birds generate and regulate body heat internally and are endothermic.

165. Which organism uses gills as the principal respiratory organ?

A. Adult frog
B. Fish
C. Lizard
D. Bird

✓ **Correct Answer: B. Fish**

Explanation: Fish use gills as their primary respiratory organs for extracting oxygen from water.

166. Which organism has a three-chambered heart under normal adult conditions?

A. Bird
B. Mammal
C. Most amphibians
D. Human

✓ **Correct Answer: C. Most amphibians**

Explanation: Most adult amphibians have two atria and one ventricle, giving a three-chambered heart.

167. Which animal has a four-chambered heart?

A. Frog
B. Fish
C. Mammal
D. Earthworm

✓ **Correct Answer: C. Mammal**

Explanation: Mammals have two atria and two ventricles, forming a four-chambered heart.

168. Which of the following represents increasing complexity of organization most appropriately?

A. Mammal → sponge → protozoan
B. Protozoan → sponge → complex multicellular animal
C. Sponge → protozoan → mammal
D. Mammal → protozoan → sponge

✓ **Correct Answer: B. Protozoan → sponge → complex multicellular animal**

Explanation: This sequence moves broadly from unicellular organization to cellular-level multicellularity and then to more complex tissue/organ-level organization.

169. Which pair is correctly matched?

A. Arthropoda — mantle
B. Mollusca — jointed appendages
C. Annelida — segmentation
D. Porifera — notochord

✓ **Correct Answer: C. Annelida — segmentation**

Explanation: Segmentation is a major annelid feature; mantle belongs to molluscs, jointed appendages to arthropods, and notochord to chordates.

170. Very twisted: An animal has bilateral symmetry, three germ layers, a true coelom, segmentation and a closed circulatory system. Which phylum is MOST likely?

A. Porifera
B. Cnidaria
C. Annelida
D. Nematoda

✓ **Correct Answer: C. Annelida**

Explanation: This combination strongly matches annelids: they are bilaterally symmetrical, triploblastic, coelomate, segmented and generally have a closed circulatory system.